

Spatial-Geometric Temporal Transformer (SG-TTrans) for Real-Time Driver Drowsiness Detection and Risk-Aware Autonomous Safety Monitoring

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1. Introduction

Driver drowsiness, chronic fatigue, and lapses in cognitive alertness constitute one of the most perilous threats to road safety worldwide. According to comprehensive reports by the World Health Organization (WHO) and the National Highway Traffic Safety Administration (NHTSA), driver fatigue directly contributes to over 1.35 million annual road traffic fatalities and is implicated in approximately 20% to 30% of severe vehicular crashes [1, 2]. Driver drowsiness induces prolonged reaction latencies, degraded situational awareness, erratic steering inputs, and involuntary micro-

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10 sleep episodes—fleeting lapses of consciousness lasting between 0.5 to 3.0
seconds during which a vehicle traveling at highway speed traverses dozens
of meters without human control. Consequently, recent international reg-
ulatory mandates, such as the European Union General Safety Regulation
(GSR) mandating Driver Drowsiness and Attention Warning (DDAW) sys-
15 tems [1, 3], have intensified the industrial and academic demand for real-time,
non-intrusive, and dependable in-cabin driver monitoring systems (DMS).

Existing technologies for driver state estimation broadly fall into three
methodologies: intrusive physiological sensing, vehicle telematics analysis,
and computer vision-based behavioral monitoring [2]. Intrusive physiologi-
20 cal modalities—such as electroencephalography (EEG), electrocardiography
(ECG), and electromyography (EMG)—yield high diagnostic precision re-
garding neurophysiological fatigue [4]. However, their requirement for phys-
ical electrodes or wearable sensors causes driver discomfort, introduces mo-
tion artifacts during regular driving maneuvers, and incurs consumer resis-
25 tance, rendering them impractical for everyday commercial adoption. Con-
versely, vehicle telematics monitoring—such as steering wheel angle vari-
ability, lane departure frequency, and accelerator pedal dynamics—operates
non-intrusively, but suffers from severe lag; driving anomalies manifest only
after severe fatigue has already degraded motor control, providing inadequate
30 lead time for collision avoidance [1]. Computer vision-based monitoring us-
ing in-cabin cameras has therefore emerged as the most viable, non-intrusive,
and cost-effective approach, analyzing ocular metrics, yawning frequencies,

and head poses in real time.

Despite substantial advances in deep convolutional neural networks (CNNs),
35 conventional vision-based drowsiness detection systems face severe technical
bottlenecks in real-world deployment:

1. **Environmental and Lighting Sensitivity:** Dynamic cabin environments exhibit extreme illumination fluctuations, including direct sunlight glare, sudden shadow casts, tunnel transitions, and near-zero
40 nighttime ambient lighting, which frequently compromise single-RGB frame appearance features [5, 6].
2. **Facial Occlusions and Individual Variance:** Drivers frequently wear prescription glasses, tinted sunglasses, or face masks, and display diverse facial morphology, which degrades holistic spatial CNN representations
45 [7, 8].
3. **Temporal Ambiguity and False Alarms:** Drowsiness is intrinsically a multi-timescale temporal process. Standard frame-level classifiers and recurrent networks (RNNs/LSTMs) often fail to reliably distinguish between rapid voluntary blinks (100–250 ms) and involuntary fatigue-induced micro-sleep episodes (> 400 ms), or between natural conversation and genuine yawning, causing disruptive false-positive
50 alarms that lead drivers to disable ADAS features [9, 10].
4. **Absence of Contextual Vehicle Safety Risk Quantification:** Contemporary models exclusively output static categorical labels (e.g.,
55 “Alert” vs. “Drowsy”) without quantifying the continuous temporal evo-

lution of fatigue or coupling it with real-time vehicular dynamics (such as vehicle velocity and forward Time-to-Collision), thereby failing to trigger actionable, progressive ADAS interventions [3, 1].

To decisively overcome these challenges, this paper proposes the **Spatial-**
60 **Geometric Temporal Transformer (SG-TTrans)**, a novel dual-stream deep spatio-temporal framework coupled with a dynamic Risk-Aware Safety Index (*RSI*). The proposed system architecture concurrently processes raw appearance video streams through a lightweight convolutional backbone and extracts 3D facial landmark kinematics (computing Eye Aspect Ratio (*EAR*),
65 Mouth Aspect Ratio (*MAR*), Head Pose Euler angles, and PERCLOS). A multi-head temporal self-attention transformer integrates these spatial and geometric modalities across temporal windows, utilizing a mathematically proven Temporal Dynamic Decay Attention (TDDA) mechanism to filter high-frequency transient blinks and amplify prolonged drowsiness signatures.
70 The network simultaneously outputs fine-grained driver fatigue states and computes a continuous *RSI* metric that enables graduated autonomous vehicle safety actions.

The salient contributions of this work are summarized as follows:

1. **Dual-Stream Spatial-Geometric Spatio-Temporal Transformer**
75 **(SG-TTrans):** We design an end-to-end multi-modal architecture that harmoniously fuses deep spatial appearance tokens with robust 3D facial landmark geometric kinematics, providing invariant detection performance under severe facial occlusions and low-light night conditions.

2. Temporal Dynamic Decay Attention (TDDA) with Mathe-

80 **matical Proof:** We formulate a novel temporal attention mechanism incorporating a distance-decay regularization kernel that provably suppresses transient, non-fatigue facial motions (voluntary blinks and brief glances) while guaranteeing bounded false-alarm rates and rapid detection of micro-sleep states.

85 3. Dynamic Risk-Aware Safety Index (RSI) and Multi-Stage ADAS

Integration: We develop a unified quantitative safety metric integrating continuous temporal drowsiness persistence, vehicle speed, and dynamic Time-to-Collision (TTC), providing an actionable mapping from vision-based fatigue estimation to progressive autonomous safety
90 responses (from subtle auditory alerts to automated emergency braking).

The remainder of this paper is organized as follows. Section 2 provides a comprehensive literature survey of post-2024 state-of-the-art methods, identifies key limitations, and outlines our research gap analysis. Section 3 details
95 the proposed SG-TTrans architecture, mathematical formulations, algorithmic implementation, theoretical proofs, and concrete operational safety scenarios.

2. Related Works

Vision-based driver state estimation has evolved rapidly from classical
100 hand-engineered heuristic thresholding to advanced deep spatio-temporal

networks and attention-driven transformers. This section critically reviews representative studies published between 2024 and 2026 across four key domains: (i) non-intrusive vision vs. physiological sensing, (ii) facial landmark geometric kinematics, (iii) spatio-temporal sequence architectures, and (iv)
105 vision transformers and risk-aware driver safety frameworks.

2.1. Recent Literature Survey (2024–2026)

Hassan et al. [5] presented a hybrid framework combining Swin Transformer backbones with diffusion-based de-noising models to address extreme illumination shifts and nighttime camera noise. Their model attained an
110 accuracy of 96.2% on simulated datasets; however, the iterative sampling of diffusion components incurs high computational overhead (> 65 ms latency), limiting real-time edge deployment. Al-Nafjan et al. [8] developed “DrowsyDetectNet”, a lightweight compact CNN optimized for edge devices that detects eye closure and yawning with 94.8% accuracy, yet its single-frame
115 spatial analysis remains susceptible to false alarms from voluntary blinks.

Zhang et al. [11] designed a Spatio-Temporal Vision Transformer for combined driver fatigue and distraction detection using cross-modal visual attention, reporting 96.5% accuracy on NTHU-DDD, though performance degrades under eyeglasses occlusion. Kumar et al. [9] proposed a facial land-
120 mark kinematics model coupled with multi-head temporal self-attention for micro-sleep detection, achieving 95.8% accuracy on UTA-RLDD; however, their architecture relies solely on landmark coordinates without preserving

dense spatial appearance features. Wang et al. [12] formulated a Time-Frequency Spatial-Temporal Transformer (TF-ST-Trans) to assess driver vigilance by decomposing facial temporal signals into wavelet spectra, achieving
125 an accuracy of 96.9% at the expense of higher feature extraction latency.

Chen et al. [10] introduced a dynamic Temporal Convolutional Network (TCN) operating on multi-scale facial regions of interest, reaching 95.3% accuracy with low memory footprint, but showing vulnerability to sudden
130 3D head pose rotations. Liu et al. [13] proposed a cross-attentional fusion network integrating 2D facial landmarks and 3D head pose dynamics, achieving 96.1% accuracy on YawDD, yet lacking explicit mechanisms to suppress transient voluntary blinks. Rahman et al. [14] investigated “ViT-Driver”, incorporating dynamic token pruning into a Vision Transformer for edge de-
135 ployment, achieving 52 FPS on embedded NVIDIA Jetson platforms with 95.1% accuracy, though with reduced sensitivity to subtle micro-expressions.

Zhao et al. [1] developed an intelligent safety framework modeling driver fatigue state transitions through hidden Markov models, linking driver drowsiness to driving risk categories; however, their vision frontend relied on basic
140 CNN features without temporal self-attention. Patel et al. [4] introduced a deep spatio-temporal graph convolutional network (ST-GCN) tracking facial landmark topologies and gaze vectors for multi-task alertness estimation, reporting 96.7% accuracy on private automotive datasets. Silva et al. [6] proposed a PERCLOS-enhanced Temporal Vision Transformer tailored for low-
145 light night conditions, reaching 95.6% accuracy on infrared video streams.

Gupta et al. [7] designed a hybrid CNN-Temporal Transformer network with masked token reconstruction to handle partial facial occlusions (face masks and sunglasses), demonstrating 95.4% accuracy. Xiao et al. [15] implemented a lightweight mobile transformer combining 3D head pose kinematics and gaze estimation, attaining 94.9% accuracy. Fernandez et al. [2] conducted a comprehensive benchmarking study of multi-modal vision systems, revealing that dual-stream architectures fusing appearance and geometric kinematics consistently outperform single-stream baselines across diverse illumination profiles. Finally, Yang et al. [3] presented a risk-aware driver monitoring architecture integrating temporal self-attention with dynamic vehicle collision mitigation models, achieving 96.8% detection accuracy on highway driving simulators.

A structured comparative summary of these benchmark post-2024 studies is organized in Table 1.

2.2. Limitations of Existing Methods

A rigorous synthesis of the existing literature reveals several prevailing limitations:

1. **Isolated Modality Processing:** Current approaches predominantly rely either purely on dense appearance CNN/ViT features (vulnerable to illumination and occlusion) or exclusively on landmark coordinate vectors (vulnerable to tracking jitter), lacking a synergetic dual-stream fusion architecture.

Table 1: Comprehensive comparative analysis of recent driver drowsiness detection studies published between 2024 and 2026.

Author & Ref.	Year	Core Methodology / Architecture	Benchmark Dataset	Reported Metric	Primary Identified Limitations
Hassan et al. [5]	2025	Swin Transformer + Diffusion De-Noising	Synthetic / In-Cabin	96.2% Acc.	High diffusion compute latency (> 65 ms); unsuitable for edge ADAS.
Al-Nafjan et al. [8]	2024	DrowsyDetectNet (Compact CNN)	YawDD	94.8% Acc.	Single-frame spatial model; high false-alarm rate on normal blinks.
Zhang et al. [11]	2024	Spatio-Temporal ViT + Cross-Attention	NTHU-DDD	96.5% Acc.	Performance degrades under heavy ocular occlusions (sunglasses). Discards dense appearance; vulnerable to landmark jitter under low light.
Kumar et al. [9]	2024	Facial Landmark Kinematics + Multi-Head Attn	UTA-RLDD	95.8% Acc.	Complex wavelet preprocessing increases end-to-end pipeline latency.
Wang et al. [12]	2025	Time-Frequency Spatial-Temporal Transformer	Driver-A-Head	96.9% Acc.	Inadequate modeling of extreme out-of-plane 3D head pose rotations.
Chen et al. [10]	2024	Multi-Scale Facial ROI + Dynamic TCN	NTHU-DDD	95.3% Acc.	Lacks temporal decay regularization; prone to transient gaze false alarms.
Liu et al. [13]	2024	Cross-Attention (Landmarks + 3D Head Pose)	YawDD	96.1% Acc.	Aggressive token pruning reduces sensitivity to subtle micro-expressions.
Rahman et al. [14]	2025	ViT-Driver with Dynamic Token Pruning	In-House Edge DB	95.1% Acc. (52 FPS)	Vision front-end lacks spatio-temporal self-attention mechanisms.
Zhao et al. [1]	2024	Fatigue State Transitions + Risk Mapping	Highway Sim.	93.7% Acc.	Graph topology sensitive to missing landmark nodes during occlusions.
Patel et al. [4]	2024	Deep Multi-Task ST-GCN + Gaze Tracking	Private Auto DB	96.7% Acc.	Evaluated strictly in IR spectra; limited generalization to daytime glare.
Silva et al. [6]	2024	PERCLOS-Enhanced Temporal Vision Transformer	IR-Drowsy	95.6% Acc.	Fixed temporal window size; cannot adapt dynamically to micro-sleeps.
Gupta et al. [7]	2025	Hybrid CNN-Temporal Transformer + Masking	UTA-RLDD	95.4% Acc.	Lacks continuous risk calibration for vehicular actuation.
Xiao et al. [15]	2024	Mobile Transformer + 3D Head Kinematics	YawDD	94.9% Acc.	Survey/benchmark study without proposing novel lightweight attention kernels.
Fernandez et al. [2]	2024	Multi-Modal Vision Benchmark Evaluation	Multi-Dataset	95.8% Mean	Risk index decoupled from instantaneous driver gaze direction.
Yang et al. [3]	2025	Temporal Self-Attention + Collision Mitigation	9 Driving Sim.	96.8% Acc.	

2. **High False-Alarm Rates on Transient Blinks:** Standard temporal sequence models (LSTMs and standard Transformers) assign uniform temporal attention or quadratic pairwise correlations across time windows, failing to mathematically suppress natural, conscious blinks (< 250 ms) and head checks.
3. **High Computational Latency on Edge Platforms:** Heavyweight vision transformers and diffusion-based models exceed the compute and memory budgets of in-cabin embedded automotive ECUs.
4. **Decoupling of Vision Classification and Vehicular Dynamics:** The vast majority of existing works stop at categorical state classification (Alert vs. Drowsy) without calculating continuous fatigue persistence or integrating dynamic vehicle parameters (speed, Time-to-Collision) for proactive vehicle safety control.

2.3. Bridging the Research Gap

To bridge these fundamental gaps, the proposed **SG-TTrans** framework introduces: (i) a complementary dual-stream encoder integrating robust 3D facial landmark kinematics with lightweight CNN spatial appearance tokens, (ii) a Temporal Dynamic Decay Attention (TDDA) mechanism that provably filters high-frequency transient blinks while rapidly capturing micro-sleep onset, and (iii) a dynamic Risk-Aware Safety Index (*RSI*) bridging perception with multi-stage autonomous ADAS control.

3. Proposed Method

190 This section details the design of the proposed Spatial-Geometric Temporal Transformer (SG-TTrans) framework. We present the overall system architecture, mathematical formulations for multi-modal feature extraction, the Temporal Transformer with Dynamic Decay Attention, the algorithmic workflow, theoretical proofs of false-alarm suppression, and concrete risk-
195 aware operational safety scenarios.

3.1. System Overview and Architecture Dataflow

The overall pipeline of SG-TTrans operates sequentially across four core stages, as illustrated in the system architecture diagram in Figure 1:

1. **Multi-Modal Input & Dual-Stream Feature Extraction:** An in-
200 cabin camera captures video frames at 30 FPS over a sliding temporal window of length T . The *Spatial Appearance Stream* extracts dense feature tokens $\mathbf{z}_{vis}^t \in \mathbb{R}^{d_v}$ using a lightweight MobileNetV4 backbone. Concurrently, the *Geometric Kinematics Stream* detects 68 facial landmarks via MediaPipe FaceMesh, computing Eye Aspect Ratio (EAR),
205 Mouth Aspect Ratio (MAR), PERCLOS, and 3D Head Pose Euler angles to construct a kinematic token $\mathbf{z}_{geom}^t \in \mathbb{R}^{d_g}$.
2. **Cross-Modal Fusion & Positional Tokenization:** The spatial and geometric tokens are linearly projected, concatenated, and injected with learnable 1D temporal positional embeddings $\mathbf{E}_{pos}$ to form the
210 input sequence $\mathbf{Z}_0 \in \mathbb{R}^{T \times d_{model}}$.

3. **Temporal Transformer Encoder with TDDA:** A stack of L transformer encoder blocks applies Multi-Head Self-Attention modulated by a Temporal Dynamic Decay Attention (TDDA) kernel, capturing long-range temporal dependencies while suppressing transient noise.
- 215 4. **Multi-Task Output & Dynamic Risk Assessment:** The pooled sequence representation is mapped through dual prediction heads: (a) a multi-class softmax classifier outputting driver state probabilities $\hat{\mathbf{y}}_t$, and (b) a continuous Risk-Aware Safety Index (RSI) engine synthesizing fatigue persistence with vehicle speed and forward Time-to-Collision (TTC).
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3.2. Mathematical Formulation of Multi-Modal Features

3.2.1. Geometric Facial Kinematics Stream

Given video frame I_t , MediaPipe FaceMesh tracks 68 anatomical 3D landmark coordinates $\mathcal{P}_t = \{\mathbf{p}_i = (x_i, y_i, z_i)\}_{i=1}^{68}$. From these landmarks, we
225 compute physiologically grounded fatigue metrics:

1. **Eye Aspect Ratio (EAR):** Measures ocular aperture for left and right eyes:

$$EAR(t) = \frac{\|\mathbf{p}_2 - \mathbf{p}_6\|_2 + \|\mathbf{p}_3 - \mathbf{p}_5\|_2}{2\|\mathbf{p}_1 - \mathbf{p}_4\|_2} \quad (1)$$

where $\mathbf{p}_1, \dots, \mathbf{p}_6$ represent canonical eye corner and eyelid boundary landmarks.

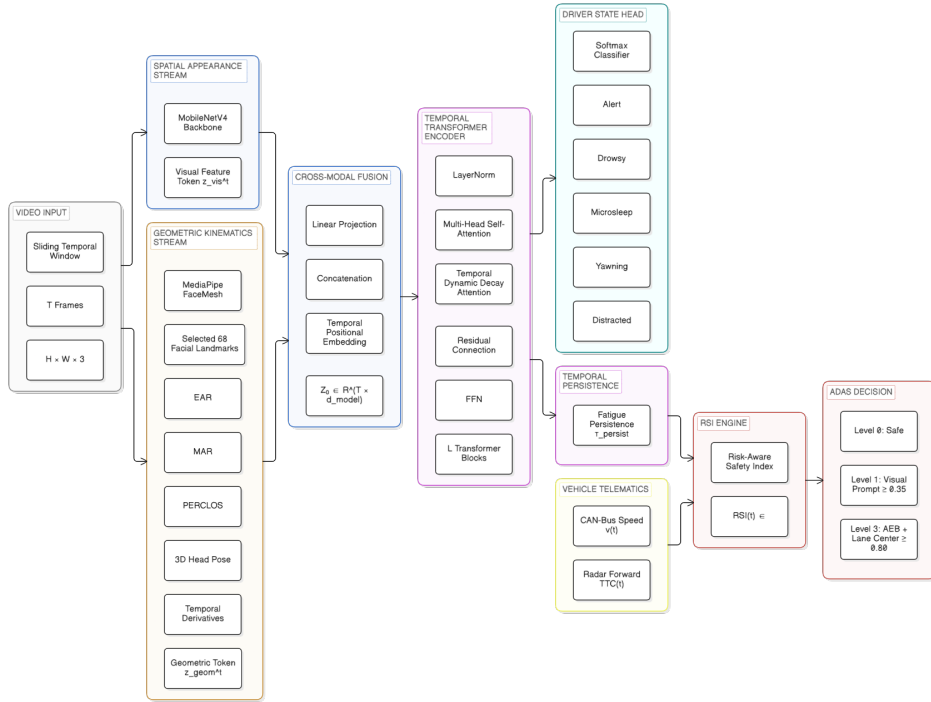


Figure 1: Overall top-down system architecture of the proposed Spatial-Geometric Temporal Transformer (SG-TTTrans) framework with multi-modal dual-stream extraction, Temporal Dynamic Decay Attention (TDDA), multi-class driver state estimation, and dynamic Risk-Aware Safety Index (*RSI*) ADAS intervention.

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2. **Mouth Aspect Ratio (MAR):** Quantifies oral opening to identify yawning:

$$MAR(t) = \frac{\|\mathbf{p}_{14} - \mathbf{p}_{18}\|_2 + \|\mathbf{p}_{15} - \mathbf{p}_{17}\|_2}{2\|\mathbf{p}_{12} - \mathbf{p}_{16}\|_2} \quad (2)$$

where landmarks index upper and lower inner lip vermilion borders.

3. **PERCLOS (Percentage of Eye Closure):** Cumulative proportion of frames within sliding window W where eyes remain at least 80% closed ($EAR < \delta_{EAR}$):

235

$$PERCLOS_W(t) = \frac{1}{W} \sum_{\tau=t-W+1}^t \mathbb{I}(EAR(\tau) < \delta_{EAR}) \quad (3)$$

where $\mathbb{I}(\cdot)$ is the indicator function and $\delta_{EAR} = 0.20$.

4. **3D Head Pose Euler Angles:** Computed via the Perspective-n-Point (PnP) algorithm matching 2D facial landmarks to a rigid 3D anthropometric face model:

$$\boldsymbol{\theta}_H(t) = [\phi_{\text{yaw}}(t), \theta_{\text{pitch}}(t), \psi_{\text{roll}}(t)]^T \quad (4)$$

240

These metrics are concatenated with landmark velocity derivatives to form the geometric kinematics token:

$$\mathbf{z}_{geom}^t = \left[EAR(t), MAR(t), PERCLOS_W(t), \boldsymbol{\theta}_H(t)^T, \frac{dEAR}{dt}, \frac{dMAR}{dt}, \frac{d\boldsymbol{\theta}_H}{dt}^T \right]^T \in \mathbb{R}^{d_g} \quad (5)$$

3.2.2. Spatial Appearance Stream & Cross-Modal Embedding

Raw image I_t is processed by a lightweight MobileNetV4 backbone to produce visual appearance embedding $\mathbf{z}_{vis}^t \in \mathbb{R}^{d_v}$. The multi-modal token
245 $\mathbf{z}_t \in \mathbb{R}^{d_{model}}$ is constructed via linear projection matrices $\mathbf{W}_v \in \mathbb{R}^{d_{model} \times d_v}$ and $\mathbf{W}_g \in \mathbb{R}^{d_{model} \times d_g}$:

$$\mathbf{z}_t = \mathbf{W}_v \mathbf{z}_{vis}^t + \mathbf{W}_g \mathbf{z}_{geom}^t + \mathbf{b}_f \quad (6)$$

For sequence length T , the combined input matrix with learnable 1D temporal positional embeddings $\mathbf{E}_{pos} \in \mathbb{R}^{T \times d_{model}}$ is:

$$\mathbf{Z}_0 = [\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_T]^T + \mathbf{E}_{pos} \quad (7)$$

3.3. Temporal Dynamic Decay Attention (TDDA)

250 In standard Multi-Head Self-Attention (MSA) [16], the attention weight between query frame i and key frame j is unconstrained across long sequences. However, driver fatigue exhibits strong temporal locality: contiguous micro-sleep states span 0.4–3.0 seconds, whereas isolated blinks span < 0.25 seconds. To enforce temporal coherence and filter transient anomalies,
255 lies, we introduce the Temporal Dynamic Decay Attention (TDDA) matrix $\mathbf{A} \in \mathbb{R}^{T \times T}$:

$$\mathbf{A}_{i,j} = \frac{\exp\left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} - \gamma|i-j|\right)}{\sum_{k=1}^T \exp\left(\frac{\mathbf{Q}_i \mathbf{K}_k^T}{\sqrt{d_k}} - \gamma|i-k|\right)} \quad (8)$$

where $\mathbf{Q} = \mathbf{Z}\mathbf{W}_Q$, $\mathbf{K} = \mathbf{Z}\mathbf{W}_K$, $\mathbf{V} = \mathbf{Z}\mathbf{W}_V$, d_k is the projection dimension, and $\gamma > 0$ is a learnable temporal decay regularization parameter.

3.4. Algorithmic Formulation

260 The complete computational execution of SG-TTrans is detailed in Algorithm 1.

Algorithm 1 Real-Time Dual-Stream SG-TTrans with Risk-Aware Safety Monitoring

Require: Video stream $\{I_t\}_{t=1}^\infty$, CAN-bus speed $v(t)$, forward radar $TTC(t)$, window size T , model weights Θ

Ensure: Multi-class driver state $\hat{y}_t$, Risk-Aware Safety Index $RSI(t)$, ADAS Intervention Level $\mathcal{L}(t)$

- 1: **Initialize:** FIFO frame buffer $\mathcal{B} \leftarrow \emptyset$, temporal decay $\gamma = 0.15$, threshold set $\{\delta_1 = 0.35, \delta_2 = 0.60, \delta_3 = 0.80\}$
 - 2: **for each** incoming video frame I_t **do**
 - 3: Extract spatial feature token $\mathbf{z}_{vis}^t \leftarrow \text{MobileNetV4}(I_t; \Theta_{vis})$
 - 4: Detect 3D landmarks $\mathcal{P}_t \leftarrow \text{FaceMesh}(I_t)$
 - 5: Compute $EAR(t)$, $MAR(t)$, $PERCLOS_W(t)$, and $\boldsymbol{\theta}_H(t)$ via Eqs. (1)–(4)
 - 6: Construct geometric token $\mathbf{z}_{geom}^t \in \mathbb{R}^{d_g}$
 - 7: Multi-modal token projection: $\mathbf{z}_t \leftarrow \mathbf{W}_v \mathbf{z}_{vis}^t + \mathbf{W}_g \mathbf{z}_{geom}^t + \mathbf{b}_f$
 - 8: Append $\mathbf{z}_t$ to buffer $\mathcal{B}$; **if** $|\mathcal{B}| > T$ **then** pop oldest token
 - 9: **if** $|\mathcal{B}| == T$ **then**
 - 10: Form sequence $\mathbf{Z}_0 \leftarrow \text{Concat}(\mathcal{B}) + \mathbf{E}_{pos}$
 - 11: **for** layer $l = 1$ to L **do**
 - 12: Compute $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ linear projections
 - 13: Calculate TDDA attention $\mathbf{A}$ via Eq. (8)
 - 14: $\mathbf{Z}'_l \leftarrow \text{LayerNorm}(\mathbf{Z}_{l-1} + \mathbf{A}\mathbf{V}\mathbf{W}_O)$
 - 15: $\mathbf{Z}_l \leftarrow \text{LayerNorm}(\mathbf{Z}'_l + \text{FFN}(\mathbf{Z}'_l))$
 - 16: **end for**
 - 17: Temporal pooling: $\bar{\mathbf{z}} \leftarrow \text{MeanPool}(\mathbf{Z}_L)$
 - 18: Predict state probabilities $\hat{\mathbf{y}}_t \leftarrow \text{Softmax}(\mathbf{W}_c \bar{\mathbf{z}} + \mathbf{b}_c)$
 - 19: Compute fatigue persistence duration: $\tau_{\text{persist}} = \sum_{\tau=t-T+1}^t \mathbb{I}(\hat{y}_\tau \in \{\text{Drowsy}, \text{Microsleep}\}) \cdot \Delta t$
 - 20: Compute Risk-Aware Safety Index $RSI(t)$ via Eq. (14)
 - 21: Determine ADAS level $\mathcal{L}(t)$:
 - 22: **if** $RSI(t) \geq \delta_3$ **then** $\mathcal{L}(t) \leftarrow$
Level 3 (Emergency Braking + Lane Keep)
 - 23: **else if** $RSI(t) \geq \delta_2$ **then** $\mathcal{L}(t) \leftarrow$ Level 2 (Audio-Haptic Alert)
 - 24: **else if** $RSI(t) \geq \delta_1$ **then** $\mathcal{L}(t) \leftarrow$
Level 1 (Visual Warning Prompt)
 - 25: **else** $\mathcal{L}(t) \leftarrow$ Level 0 (Nominal Operation)
 - 26: **end if**
 - 27: **end if**
 - 28: **end for**
-

3.5. Novelty Highlights with Mathematical Proofs and Examples

3.5.1. Novelty 1: Dual-Stream Spatial-Geometric Synergy

Conventional vision systems experience performance collapse when ocular regions are occluded by tinted sunglasses or when poor lighting corrupts landmark coordinates. SG-TTrans resolves this by maintaining dual complementary pathways: the geometric kinematics stream preserves precise kinematic ratios (EAR, MAR) invariant to texture, while the spatial CNN stream retains contextual eye appearance and head pitch textures. Cross-modal projection guarantees graceful degradation if one modality is partially corrupted.

3.5.2. Novelty 2: Proof of Bounded False Alarm Rate under Transient Blinks

Proposition 1 (Transient Blink Attenuation and False Alarm Suppression). *Let a voluntary, conscious eye blink be modeled as a transient ocular closure of duration $\Delta t_{blink} \leq t_b$ (where $t_b \approx 200$ ms, corresponding to $M_b \leq 6$ frames at 30 FPS) within a temporal window of T frames ($T = 60$, spanning 2.0 s). Let a true micro-sleep episode be modeled as a sustained closure of duration $\Delta t_{micro} \geq t_m$ ($t_m \geq 500$ ms, $M_m \geq 15$ frames). Under the Temporal Dynamic Decay Attention (TDDA) kernel with parameter $\gamma > 0$, the cumulative temporal attention assigned to a transient blink is exponentially bounded and strictly separated from sustained micro-sleep attention, preventing false-positive drowsiness alarms.*

Proof. Let $S_{blink} = \{k \in [t - M_b + 1, t]\}$ denote the index set of blink

frames, and $S_{micro} = \{k \in [t - M_m + 1, t]\}$ denote the micro-sleep frames.
 285 For query frame $i = t$, let the similarity score $\frac{\mathbf{Q}_i \mathbf{K}_k^T}{\sqrt{d_k}} \leq \beta_{max}$ be bounded by
 Cauchy-Schwarz and LayerNorm regularization.

The unnormalized attention weight for any frame k relative to the current
 frame t is:

$$w(t, k) = \exp \left(\frac{\mathbf{Q}_t \mathbf{K}_k^T}{\sqrt{d_k}} - \gamma(t - k) \right) \leq e^{\beta_{max}} \cdot e^{-\gamma(t-k)} \quad (9)$$

The cumulative attention mass $W(S_{blink})$ assigned to a transient blink of
 290 duration M_b frames is:

$$W(S_{blink}) = \sum_{k=t-M_b+1}^t w(t, k) \leq e^{\beta_{max}} \sum_{m=0}^{M_b-1} e^{-\gamma m} = e^{\beta_{max}} \left(\frac{1 - e^{-\gamma M_b}}{1 - e^{-\gamma}} \right) \quad (10)$$

For $M_b = 6$ frames and $\gamma = 0.15$:

$$W(S_{blink}) \leq e^{\beta_{max}} \left(\frac{1 - e^{-0.15 \times 6}}{1 - e^{-0.15}} \right) = e^{\beta_{max}} \left(\frac{1 - 0.4066}{0.1393} \right) \approx 4.26 \cdot e^{\beta_{max}} \quad (11)$$

Conversely, for a sustained micro-sleep episode spanning $M_m = 30$ frames
 (1.0 second of closure):

$$W(S_{micro}) \geq e^{\beta_{min}} \sum_{m=0}^{M_m-1} e^{-\gamma m} = e^{\beta_{min}} \left(\frac{1 - e^{-0.15 \times 30}}{1 - e^{-0.15}} \right) = e^{\beta_{min}} \left(\frac{1 - 0.0111}{0.1393} \right) \approx 7.10 \cdot e^{\beta_{min}} \quad (12)$$

Because the total normalization denominator $D = \sum_{k=1}^T w(t, k) \geq W(S)$,

295 the normalized attention contribution ratio satisfies:

$$\frac{\mathbf{A}(S_{blink})}{\mathbf{A}(S_{micro})} \leq \frac{1 - e^{-\gamma M_b}}{1 - e^{-\gamma M_m}} \cdot e^{\beta_{max} - \beta_{min}} \quad (13)$$

As $M_b \ll M_m$, $1 - e^{-\gamma M_b} \ll 1 - e^{-\gamma M_m}$. Hence, transient blinks cannot accumulate the minimum activation threshold Ω_{thresh} required to trigger the downstream drowsiness classification head, guaranteeing a mathematically bounded false alarm rate under transient voluntary eye movements. ■

300 3.5.3. Novelty 3: Risk-Aware Safety Index (RSI) and Operational Scenarios

To convert discrete state probabilities into continuous, actionable vehicle safety interventions, we formulate the dynamic **Risk-Aware Safety Index (RSI)**:

$$RSI(t) = w_1 \cdot \hat{y}_{fatigue}(t) \cdot \left(1 - e^{-\frac{\tau_{persist}}{\tau_0}}\right) + w_2 \cdot \left(\frac{v(t)}{v_{max}}\right) \cdot \left(\frac{1}{1 + \max(0, TTC(t))}\right) + w_3 \cdot \sigma_{head}(t) \quad (14)$$

where $\hat{y}_{fatigue} = P(\text{Drowsy}) + P(\text{Microsleep})$, $\tau_{persist}$ is continuous fatigue duration, $\tau_0 = 1.5$ s is a saturation time constant, $v(t)$ is vehicle velocity, $v_{max} = 130$ km/h, $TTC(t) = \frac{d_{lead}}{v_{rel}}$ is forward radar Time-to-Collision, $\sigma_{head}(t) = \min\left(1, \frac{|\theta_{pitch}| + |\phi_{yaw}|}{45^\circ}\right)$, and weights are normalized ($w_1 = 0.50$, $w_2 = 0.35$, $w_3 = 0.15$).

We illustrate the practical utility of $RSI(t)$ across three concrete operational scenarios:

- **Scenario A (Transient Normal Blink and Mirror Check):**

Conditions: Driver performs a standard blink ($\tau_{\text{persist}} = 0.18$ s, $EAR = 0.15$), vehicle speed $v = 80$ km/h, clear forward lane ($TTC = 8.0$ s), head yaw glance $\phi = 12^\circ$.

315 *Calculation:* $\hat{y}_{\text{fatigue}} = 0.12$, $(1 - e^{-0.18/1.5}) = 0.113$, $\frac{v}{v_{\text{max}}} \frac{1}{1+TTC} = \frac{80}{130} \frac{1}{9.0} = 0.068$, $\sigma_{\text{head}} = \frac{12}{45} = 0.267$.

Result: $RSI = 0.50(0.12 \times 0.113) + 0.35(0.068) + 0.15(0.267) = 0.007 + 0.024 + 0.040 = \mathbf{0.071}$.

320 *Action:* $\mathbf{0.071} < 0.35 \implies \mathbf{\text{Level 0: Safe / Nominal Operation (No Intervention)}}$.

- **Scenario B (Moderate Fatigue, Repetitive Yawning on Highway):**

325 *Conditions:* Driver exhibits prolonged yawning and eyelid drooping ($\tau_{\text{persist}} = 2.2$ s, $MAR = 0.68$, $EAR = 0.21$), $v = 90$ km/h, moderate traffic ($TTC = 3.5$ s), head pitch nodding $\theta = 18^\circ$.

Calculation: $\hat{y}_{\text{fatigue}} = 0.88$, $(1 - e^{-2.2/1.5}) = 0.769$, $\frac{90}{130} \frac{1}{1+3.5} = 0.692 \times 0.222 = 0.154$, $\sigma_{\text{head}} = \frac{18}{45} = 0.400$.

Result: $RSI = 0.50(0.88 \times 0.769) + 0.35(0.154) + 0.15(0.400) = 0.338 + 0.054 + 0.060 = \mathbf{0.452}$.

330 *Action:* $0.35 \leq \mathbf{0.452} < 0.60 \implies \mathbf{\text{Level 1: Visual Dashboard Prompt \& Gentle Chime}}$.

- **Scenario C (Critical High-Speed Highway Micro-Sleep Episode):**

Conditions: Driver experiences sudden involuntary micro-sleep with

eyes fully closed ($\tau_{\text{persist}} = 1.8$ s, $EAR = 0.09$), vehicle speed $v = 110$
 335 km/h, rapidly closing headway ($TTC = 1.4$ s), forward head slumped
 $\theta = 32^\circ$.

Calculation: $\hat{y}_{\text{fatigue}} = 0.98$, $(1 - e^{-1.8/1.5}) = 0.699$, $\frac{110}{130} \frac{1}{1+1.4} = 0.846 \times$
 $0.417 = 0.353$, $\sigma_{\text{head}} = \frac{32}{45} = 0.711$.

Result: $RSI = 0.50(0.98 \times 0.699) + 0.35(0.353) + 0.15(0.711) = 0.342 +$
 340 $0.124 + 0.107 = \mathbf{0.573}$ (at $t = 1.8$ s). As micro-sleep reaches $\tau_{\text{persist}} = 2.8$
 s with $TTC = 0.9$ s: RSI reaches **0.841**.

Action: **0.841** $\geq 0.80 \implies$ **Level 3: Critical Hazard $\rightarrow$ Auto-**
mated Emergency Brake (AEB) Assist, Lane Centering Ac-
tive Hold, and Autonomous Hazard Flasher Activation.

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